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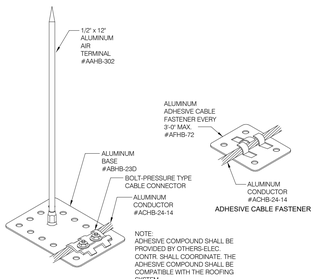
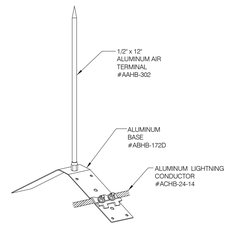
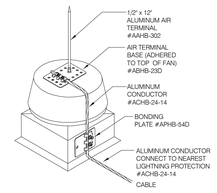


1 / E8.2H AIR TERMINAL BASE
SCALE: NTS
SYMBOL *A

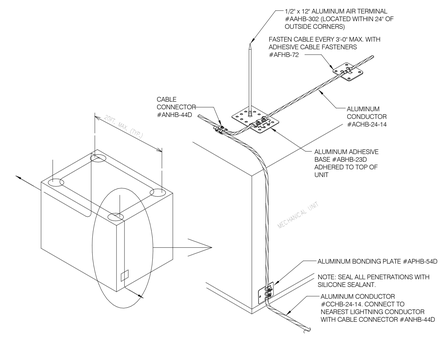


2 / E8.2H SADDLE AIR TERMINAL BASE
SCALE: NTS
SYMBOL *S

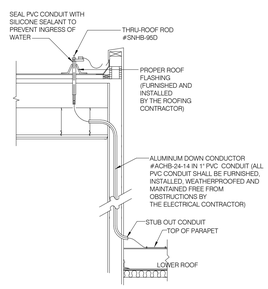
NOTE: AIR TERMINALS ARE REQUIRED ON ROOF VENTILATORS/EXH. FANS THAT ARE NOT WITHIN A ZONE OF PROTECTION OR WHEN HIGHER THAN ADJACENT AIR TERMINALS.



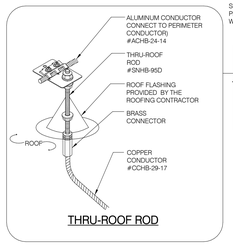
3 / E8.2H AIR TERMINAL AT EXH.FAN
SCALE: NTS



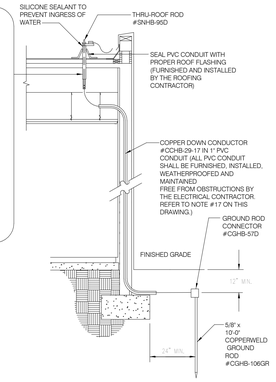
4 / E8.2H LIGHTENING PROTECTION AT ROOF TOP UNIT.
SCALE: NTS



5 / E8.2H DOWN CONDUCTOR TO LOWER ROOF
SCALE: NTS
SYMBOL



THRU-ROOF ROD



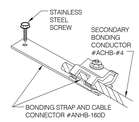
6 / E8.2H LIGHTENING PROTECTION GROUNDING
SCALE: NTS
SYMBOL

LIGHTNING PROTECTION NOTES

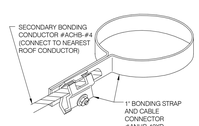
1. ALL MATERIALS SHOWN ARE MANUFACTURED BY HEARY BROS. LIGHTNING PROTECTION CO., INC.
2. THE LIGHTNING PROTECTION SYSTEM AS SHOWN ON DRAWING HAS BEEN DESIGNED IN ACCORDANCE WITH UL96 & NFPA-780 LIGHTNING PROTECTION SYSTEM STANDARDS.
3. CONDUCTORS SHALL MAINTAIN A HORIZONTAL OR DOWNWARD COURSE, FREE FROM "U" OR "V" (DOWN AND UP) POCKETS.
4. NO BEND OF CONDUCTOR SHALL FORM AN ANGLE OF LESS THAN 90° NOR SHALL HAVE A RADIUS OF BEND LESS THAN 8".
5. AIR TERMINALS SHALL BE SPACED EVERY 20'-0" MAXIMUM AROUND THE ROOF PERIMETER AND/OR ALONG ROOF RIDGES. AIR TERMINALS SHALL BE LOCATED WITHIN 2'-0" OF OUTSIDE CORNERS.
6. AIR TERMINALS SHALL BE SPACED EVERY 50'-0" MAXIMUM IN CENTER ROOF AREAS.
7. ACTUAL JOBSITE CONDITIONS MAY REQUIRE SLIGHT ALTERATIONS IN AIR TERMINAL, DOWN CONDUCTOR AND GROUND ROD LOCATIONS.
8. BARE COPPER MATERIALS SHALL NOT BE INSTALLED ON ALUMINUM OR GALVALUM SURFACES, AND ALUMINUM MATERIALS SHALL NOT BE INSTALLED ON COPPER SURFACES.
9. ALL LIGHTNING PROTECTION CONDUCTORS SHALL BE FASTENED EVERY 3'-0" MAX.
10. METALLIC BODIES OF INDUCTANCE SITUATED WITHIN 6'-0" OF A LIGHTNING CONDUCTOR OR ANOTHER BONDED METAL BODY SHALL BE INTERCONNECTED TO THE LIGHTNING CONDUCTOR SYSTEM, UNLESS INHERENTLY GROUNDED.
11. BOND TO ALL METAL BODIES OF CONDUCTANCE WITHIN 6'-0" OF THE MAIN LIGHTNING CONDUCTOR SUCH AS EXHAUST FANS, ROOF VENTS, METAL COOLING TOWERS, H.V.A.C. UNITS, LADDERS, RAILINGS, ANTENNAS, SKYLIGHTS, METAL STACKS AND ANY OTHERS LARGE METAL BODY WHOSE HEIGHT EXCEEDS THAT OF THE AIR TERMINAL IN USE, UNLESS PROTECTED BY HIGHER ROOF ELEVATIONS.
12. CONNECTIONS TO GROUND RODS SHALL BE MADE AT A POINT NOT LESS THAN 1'-0" BELOW FINISHED GRADE AND 2'-0" AWAY FROM FOUNDATION WALL.
13. BOND TO WATERLINES (DOMESTIC & FIRE).
14. A LIGHTNING ARRESTOR, PROTECTOR OR ANTENNA DISCHARGE UNIT SHALL BE INSTALLED ON EACH ELECTRIC AND TELEPHONE SERVICE AND RADIO AND TELEVISION ANTENNA LEAD-IN BY THE ELECTRICAL CONTRACTOR, IN ACCORDANCE WITH NFPA-70.
15. TRANSIENT VOLTAGE SURGE SUPPRESSION (TVSS) OF SERVICES SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR. (I.E. COMPUTERS, COPIERS, TELEPHONE, ETC.).
16. UPON COMPLETION OF INSTALLATION UL MASTER LABEL SHALL BE ISSUED.
17. CONTRACTOR TO COORDINATE WITH ICF WALL SYSTEM. CONDUCTOR MUST RUN THROUGH INTERIOR SIDE INSULATION OF ICF SYSTEM AND BE COORDINATED WITH OTHER CONDUIT IN WALL. PENETRATION THROUGH CONCRETE WALL BELOW GRADE MUST BE COORDINATED PRIOR TO POUR OF ICF WALL SYSTEM.

LIGHTNING PROTECTION MATERIALS LIST

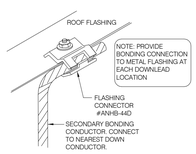
	ALUMINUM LIGHTNING PROTECTION MAIN CONDUCTOR	ACHB-24-14
	ALUMINUM LIGHTNING PROTECTION SECONDARY BONDING CABLE*	ACHB-#4
	COPPER LIGHTNING PROTECTION DOWN CONDUCTOR	CCHB-29-17
	BIMETAL CONNECTOR	MNHB-93
	CABLE FASTENERS (FASTEN CABLE EVERY 3FT. MAX.)	AFHB-72, 66, 64, CFHB-64
* A	1/2" x 12" ALUMINUM AIR TERMINAL AND ADHESIVE BASE	AAHB-302, ABHB-23D
	1/2" x 12" ALUMINUM AIR TERMINAL AND ADHESIVE BASE (AT ROOF TOP EQUIPMENT)	AAHB-302, ABHB-23D
* S	1/2" x 12" ALUMINUM AIR TERMINAL AND SADDLE BASE	AAHB-302, ABHB-172D
△	THRU-ROOF ROD	SNHB-95D
	SECONDARY BONDING: *FLASHING CONNECTOR *METAL ROOF DRAIN / GUTTER CONNECTOR *METAL VENT PIPE CONNECTOR	 ANHB-44D ANHB-160D ANHB-42XD
	ALUMINUM BONDING PLATE (AT ALUM. RTU & FANS)	APHB-54D
	CORROSION RESISTANT COPPER BONDING PLATE (TO BASE OF STEEL AT EACH DOWNLEAD)	PPHB-54D
	PIPE CLAMP (ANTENNAS, RAILINGS, ETC.)	PMHB-103D
	"C" CLAMP (LADDERS)	PPHB-54D
	CABLE CONNECTOR	ANHB-44D, CNHB-44D
	STRAIGHT SPLICER	ARHB-26
	CROSSOVER CABLE CONNECTOR	CMHB-57XXX
	WATERLINE CONNECTOR (FIRE WATER & DOMESTIC WATER)	CMHB-97D
	5/8" x 10'-0" COPPERWELD GROUND ROD AND CONNECTOR	CGHB-106GR, CGHB-57D



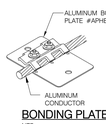
BONDING LUG (GUTTERS)
NTS



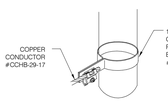
BONDING STRAP AT (VENT PIPES)
NTS



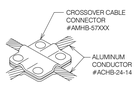
FLASHING CONNECTOR
NTS



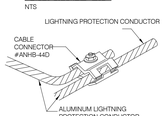
BONDING PLATE
NTS



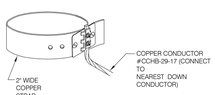
PIPE CLAMP (RAILINGS, ANTENNAS, ETC.)
NTS



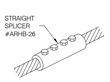
CROSSOVER CABLE CONNECTOR
NTS



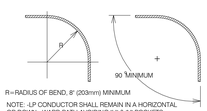
CABLE CONNECTOR
NTS



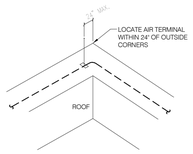
WATERLINE BONDING STRAP
NTS



STRAIGHT SPLICER
NTS



8" MIN. RADIUS BEND IN CABLE



AIR TERMINALS SHALL BE LOCATED WITHIN 24" OF OUTSIDE CORNERS
NTS

SETTY
ONE SOUTH STREET, SUITE 1110
BALTIMORE, MARYLAND 21202
TEL 667-329-5035
WATER RESOURCES MANAGEMENT · FLOOD MANAGEMENT
ENERGY MANAGEMENT SERVICES · DESIGN BUILD SERVICES
CONSTRUCTION MANAGEMENT · CONSTRUCTION SERVICES
DISTRICT OF COLUMBIA · MARYLAND
www.SETTY.com

PROJECT NUMBER: SA2017-17 # # # #

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GRIMM AND PARKER

GP# 21553

LIGHTNING PROTECTION DETAILS
HOLABIRD ACADEMY PK-8
CITY OF BALTIMORE, MARYLAND

DATE	DESCRIPTION

E-8.2H
03/13/2017
BID SET

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